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Multi-species occupancy models as robust estimators of community richness

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Abstract

1. Understanding patterns of diversity is central to ecology and conservation, yet estimates of diversity are often biased by imperfect detection. In recent years, multi-species occupancy models (MSOM) have been developed as a statistical tool to account for species-specific heterogeneity in detection while estimating true measures of diversity. Although the power of these models has been tested in various ways, their ability to estimate gamma diversity—or true community size, *N* is a largely unrecognized feature that needs rigorous evaluation.
2. We use both simulations and an empirical dataset to evaluate the bias, precision, accuracy and coverage of estimates of *N* from MSOM compared to the widely applied iChao2 non-parametric estimator. We simulated 5,600 datasets across seven scenarios of varying average occupancy and detectability covariates, as well as varying numbers of sites, replicates and true community size. Additionally, we use a real dataset of surveys over 9 years (where species accumulation reached an asymptote, indicating true *N*), to estimate *N* from each annual survey.
3. Simulations showed that both MSOM and iChao2 estimators are generally accurate (i.e. unbiased and precise) except under unideal scenarios where mean species occupancy is low. In such scenarios, MSOM frequently overestimated *N*. Across all scenarios, MSOM estimates were less certain than iChao2, but this led to over-confident iChao2 estimates that showed poor coverage. Results from the real dataset largely confirmed the simulation findings, with MSOM estimates showing greater accuracy and coverage than iChao2.
4. Community ecologists have a wide choice of analytical methods, and both iChao2 and MSOM estimates of *N* are substantially preferable to raw species counts. The simplicity of non-parametric estimators has obvious advantages, but our results show that in many cases, MSOM may provide superior estimates that also account more accurately for uncertainty. Both methods can show strong bias when average occupancy is very low, and practitioners should show caution when using estimates derived from either method under such conditions.

KEYWORDS

Chao estimator, community richness, data augmentation, gamma diversity, imperfect detection, multi-species occupancy models, occupancy modelling

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1 | INTRODUCTION

Understanding spatial and temporal patterns of biological richness (e.g. genotypic, species, functional) is central to ecology, evolutionary biology and conservation (Gotelli & Colwell, 2001). However, richness is often impossible to observe in nature because many types are rare and therefore go undetected during surveys, especially in diverse communities (Gotelli & Colwell, 2001). Even long-term monitoring projects discover previously undetected species after decades of intensive monitoring (Longino, Coddington, & Colwell, 2002). This challenge limits our ability to test key theories explaining patterns of spatial and temporal richness and could prevent effective conservation of biodiversity (Brooks, Fonseca, & Rodrigues, 2004; Ehrlich, 1996). Consequently, there is a long history of trying to estimate true richness from the number of observed types in a survey area (Fisher, Corbet, & Williams, 1943; Toulmin & Good, 1956).

Recent statistical advances in ecology have led to a suite of models—broadly referred to as occupancy models (MacKenzie et al., 2006)—which aim to correct for imperfect detection of individual species through the hierarchical modelling of both an underlying true state process (e.g. occurrence) and an intervening detection process through which the observed state is filtered. While occupancy models originated as single-species formulations (MacKenzie et al., 2002), they were quickly adapted into multi-species settings, using hierarchical random effects structures on slope and intercept parameters to relate many different species into a single 'community' or 'multi-species' occupancy model (Dorazio & Royle, 2005; Dorazio, Royle, Soderstrom, & Glimskar, 2006). In recent years, multi-species occupancy models (MSOM) have been used in a wide variety of settings to understand community-level occurrence patterns, estimate local species richness and community turnover, and robustly evaluate community assembly hypotheses (Broms, Hooten, & Fitzpatrick, 2015; Iknayan, Tingley, Furnas, & Beissinger, 2014; Jarzyna & Jetz, 2016). Despite the potential for MSOM to re-shape community ecology methods while accounting for imperfect detection, many methodological aspects of how these models are used in community ecology have not been broadly evaluated.

One key statistical side-note in the original formulation of MSOM concerned the use of 'data augmentation' approaches to estimate the true number of species in a regional community (hereafter *N*), accounting for both species that were observed at least once somewhere and unidentified species that were never observed at any site (Dorazio & Royle, 2005). Thus, *N* from MSOM can be used to provide estimates of regional community size, meta-community size and/or gamma diversity (Iknayan et al., 2014). MSOM are increasingly being used to estimate these ecological quantities (reviewed in Guillera-Arroita, Kéry, & Lahoz-Monfort, 2019), yet questions remain about the accuracy or bias of the estimates (Guillera-Arroita et al., 2019), and how appropriate they are relative to other, simpler, richness estimators such as non-parametric methods (Iknayan et al., 2014).

Fundamentally, both MSOM and non-parametric richness estimators address the same question of how many species occur in a given area. Inherent in this challenge is the question of how many species were present but not observed during sampling (Colwell et al., 2012). Non-parametric richness estimators arose from the same mathematical background of mark–recapture models as MSOM (Colwell & Coddington, 1994; MacKenzie et al., 2002), but require no assumption on the nature of species' detection probabilities. Among the most commonly used non-parametric estimators are Chao (Chao, 1984), Jackknife (Burnham & Overton, 1978, 1979; Colwell & Coddington, 1994; Heltshe & Forrester, 1983), and Bootstrap (Colwell & Coddington, 1994; Smith & van Belle, 1984). All are based around the idea that more observed rare species in a community equals more unobserved species in the true community. Chao, particularly iChao2, is the simplest to calculate (Colwell et al., 2012), is a lower bound estimator—meaning that it is designed to estimate the minimum number of unobserved species (Chao, 1984)—and has been shown to be unbiased when certain detection probabilities of rare species are homogenous (Chao & Colwell, 2017).

In this study, we directly compare *N* estimated using MSOM with *N* estimated using an analogous and common non-parametric method, iChao2, with the goal of contrasting the bias, precision, accuracy and coverage probability (Walther & Moore, 2005) of each method under various scenarios. As recently developed models that can also provide inference on other aspects of community structure, MSOMs may be attractive to many potential users, yet the models have higher data requirements, can be computationally demanding and non-intuitive to fit, and contain numerous assumptions that may not be met by the data (Guillera-Arroita et al., 2019; Iknayan et al., 2014; Rota, Fletcher, Dorazio, & Betts, 2009). Non-parametric estimators such as iChao2, in contrast, are simple to calculate, are flexible for many situations, and have relatively fewer assumptions. Consequently, if non-parametric estimates are equally as reliable as MSOM estimates in a given situation, then ecologists may prefer to use the simpler non-parametric methods. Aside from one study by Guillera-Arroita et al. (2019)—which focuses on the strength of MSOM estimates under non-ideal sampling situations and sensitivity to assumption violations and model form—MSOM estimates of *N* have not been rigorously evaluated, and there is a particular need for direct comparisons between estimates from MSOMs and other methods.

To evaluate MSOM estimates of *N* relative to non-parametric estimates, we use two approaches: simulation and empirical data. In simulations, we are able to vary the true community size (from small to large) and evaluate estimator performance while varying detectability (high to low mean, high to low variability) and occupancy (high to low mean, high to low variability) across the community. We additionally take advantage of a long-term, high-effort monitoring program of birds inhabiting recent post-fire forests. In this system, with spatially extensive annual surveys conducted from 2010 to 2018, species accumulation has plateaued; thus, the true community richness is essentially known. Consequently, this allows us to estimate *N* from annual slices of the dataset and to compare these annual estimates to the asymptotic richness observed across nearly a decade of surveys.

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2 | MATERIALS AND METHODS

2.1 | Simulating imperfectly observed communities

We simulated communities and surveys with a suite of different parameters to determine how variation in underlying community characteristics and study designs affect estimates of *N*. We chose values for each variable of interest based on a range of values observed in previous studies that used MSOMs to estimate *N* (Guillera-Arroita et al., 2019). We thus varied the number of survey sites (30 or 60 sites), number of replicate surveys at each survey site (2 or 7 replicates), true community size (20 or 125 species) and the mean and variance of detection and occupancy probability of all species in the community (i.e. occupancy and detection hyper-parameters; Table 1). We limited the number of potential variable combinations by selecting seven scenarios of detection and occupancy probability hyper-parameters (Table 1). The seven combinations included an ideal scenario (i.e. mean detection and occupancy probability of species in the community were high and there was little variation among species) and two unideal scenarios (i.e. mean detection and occupancy probability of species in the community were low and variance among species was either high or low). The remaining four combinations allowed us to evaluate how changing a single detection or occupancy probability parameter affected estimates of community size, relative to the ideal scenario (Table 1).

We simulated 100 replicate datasets for each combination (n = 56) of community and survey variables, including the true community size (N). We limited the number of replicate datasets to 100 per combination due to the computational demands of fitting Bayesian occupancy models (approximately 8 min per simulation). To simulate a dataset, we first assigned individual detection (p_i) and occupancy (ψ_i) probabilities to each species (i) in the community, as randomly drawn from normal distributions defined by the hyper-parameters in Table 1. We then created a binary matrix of true occupancy (z, with a maximum of N species) for each species (i) at each site (j), where occupied sites were indicated with a 1 and unoccupied sites were indicated with a 0. We randomly assigned true occupancy of each species at each site using:

$$z_{i,j} \sim \text{Bern}(\psi_i).$$

We simulated replicate surveys at each site using:

$$y_{i,j} \sim \text{Binomial}(n_{\text{reps}}, p_i \times z_{i,j}),$$

where n_{reps} is the number of replicate surveys at each site and $y_{i,j}$ is the number of replicate surveys where species i was detected at site j. Setting the probability of the Binomial distribution to $p_i \times z_{i,j}$ ensures that species can only be detected at sites that are truly occupied (MacKenzie et al., 2002). Given that $y_{i,j}$ was simulated directly given imperfect detection from the true occurrence matrix, $z_{i,j}$, the total number of observed species (N_{obs}) was often much less than the dimensionality of the matrix (e.g. from a true community size of 20, only 10 species might be observed at least once across all simulated site histories).

Additionally, we conducted a second set of simulations to specifically understand how mean occupancy probability of species in the community (μ_ψ) affected estimates of community size. We simulated 100 replicate datasets for each of nine different μ_ψ values spaced evenly between 0.1 and 0.9. In all of these simulations, we simulated 50 survey sites, three replicate surveys at each survey site and 50 species in the community. We randomly assigned detection probability of each species in the community from a normal distribution with a mean of 0.5 on the probability scale and standard deviation of 0.5 on the logit scale. Otherwise the simulations and analysis were identical to that described above.

2.2 | Estimating community size from simulated data

We used the simulated survey data to estimate the true community size using two methods: (a) an MSOM, and (b) an improved incidence-based Chao estimator, iChao2 (Chiu, Wang, Walther, & Chao, 2014). We fit the MSOM in JAGS (Plummer, 2003) using the data augmentation approach (Iknayan et al., 2014; Royle & Dorazio, 2010). Data

TABLE 1 Combinations of parameters used to define seven different community types based on the distribution of occupancy and detection probability among species in the community. For each scenario, occupancy and detectability values were drawn from normal distributions on the logit-scale, with means centred at the inverse logit-transformed value of the number reported below, and standard deviations (*SD*) as reported below

Community type	Average occupancy probability	*SD* occupancy probability	Average detection probability	*SD* detection probability	Description
1	0.7	0.2	0.7	0.2	Ideal scenario
2	0.7	1.0	0.7	0.2	High variation in occupancy probability
3	0.1	0.2	0.7	0.2	Low average occupancy probability
4	0.7	0.2	0.7	1.0	High variation in detection probability
5	0.7	0.2	0.3	0.2	Low average detection probability
6	0.1	1.0	0.3	1.0	Unideal scenario
7	0.1	0.2	0.3	0.2	Unideal scenario

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augmentation appends an arbitrarily large number of unobserved species to the end of the *y*-matrix described above and then estimates the proportion of all species (observed and unobserved) that are truly present in the community (Ω). We used the following hierarchical model to estimate the true number of species in the community (*N*):

$$N = \Omega \times \left(N_{\text{obs}} + N_{\text{aug}}\right),$$

$$w_i \sim \text{Bernoulli}\left(\Omega\right),$$

$$z_{i,j} \sim \text{Bernoulli}\left(w_i \times \psi_i\right),$$

$$y_{i,j} \sim \text{Binomial}\left(n_{\text{reps}}, z_{i,j} \times p_i\right),$$

where N_{obs} is the number of observed species, N_{aug} is the number of augmented species, and w_i represents a latent variable indicating whether species *i* is present in the community. We set $N_{\text{aug}} = N_{\text{obs}}$, as N_{aug} should be an arbitrarily large number such that the posterior estimate of Ω is not constrained at 1 (Royle & Dorazio, 2008). We modelled ψ_i and p_i as random effects governed by scenario-specific hyper-parameters:

$$\text{logit}\left(\psi_i\right) = \beta_i,$$

$$\beta_i \sim \text{Normal}\left(\mu_\psi, \sigma_\psi\right),$$

$$\text{logit}\left(p_i\right) = \alpha_i,$$

$$\alpha_i \sim \text{Normal}\left(\mu_p, \sigma_p\right).$$

We used vague priors for the mean and precision (mean: normal with $\mu = 0$, $\tau = 0.001$; precision: gamma with $r = .1$, $\lambda = 0.1$), and a flat uniform prior (0, 1) for Ω. We used three chains with 20,000 MCMC iterations, a burn-in of 1,000 iterations, and thinned by selecting every 25th iteration. Preliminary tests suggested that these MCMC parameters provided suitable chain convergence as assessed visually and by evaluating whether the Gelman–Rubin statistic was <1.1 (Gelman, Carlin, Stern, & Rubin, 2004).

The iChao2 estimator does not require replicate surveys (Chiu et al., 2014). Therefore, we calculated iChao2 in two ways: (a) we randomly chose a replicate survey to represent the situation when only one survey replicate was completed at each site, and (b) we pooled the results from all replicate surveys such that a site was considered occupied by a species if the species was detected on any one of the multiple replicate surveys. The first method, although not a fair comparison (as information is discarded), simulates the often frequent choice of study designs to forego repeat sampling on species occurrences. We calculated iChao2 using the 'SPADER' package (Chao, Ma, Hsieh, & Chiu, 2016).

2.3 | Evaluating community size estimators

We evaluated both estimators using four criteria: (a) bias, (b) precision, (c) accuracy and (d) coverage probability (Walther & Moore, 2005). We calculated error as the estimate of community size minus the true community size. We used the posterior mean of $\Omega \times \left(N_{\text{obs}} + N_{\text{aug}}\right)$ as a point estimate of community size from the MSOM. For each simulated scenario, we measured bias as the mean error from the 100 replicate simulations divided by the true community size (i.e. scaled mean error). We measured precision as the standard deviation of the community size estimates divided by the mean estimate from the 100 replicate simulations (i.e. coefficient of variation). We measured accuracy as the root-mean-squared error from the 100 replicate simulations divided by the true community size (i.e. scaled root-mean-squared error). Last, we measured coverage probability by calculating the proportion of simulations where the 95% credible (MSOM) or confidence interval (iChao2) contained the true value.

In many simulated communities, there were no singletons or doubletons, which are required for estimating community size using generalized Chao methods (Chiu et al., 2014). In these cases the 'SPADER' package returned an error. When evaluating bias, precision and accuracy for these cases, we thus set the estimated number of species in the community equal to the observed number of species. However, we excluded these cases when evaluating coverage probability and did not evaluate coverage probability for any scenario where more than 90 of the 100 simulations had no singletons or doubletons.

2.4 | Application of diversity estimators to real data

As a case study, we applied iChao2 and MSOM estimators of *N* to an annual dataset of avian community surveys from 2010 to 2018 conducted by The Institute for Bird Populations as part of a bioregional monitoring program in burned conifer forests. This study was designed to annually characterize the full community of terrestrial breeding birds inhabiting recent (≤10 years) post-fire areas on public lands within the Sierra Nevada and southern Cascade ecoregions of California. In each year, a revolving random panel of ~50 forests that burned within the last decade were visited, and at each, a median of nine point counts were conducted. More information on these surveys can be found in Appendix S1.

Despite extensive survey effort (~400 survey points per year), each year missed detecting a substantial number of species found in post-fire forests in other years. Surveys identified between 78 (in 2010) and 99 (in 2012) terrestrial bird species per year, with an average of 88 species. Across all years, 123 species were recorded, indicating that in any given year, an average of 28% of the regional avian community was missed. Over time, however, the accumulation of new species to the dataset slowed (Figure S1). Between 2012 and 2017, only seven new species were added, with no new species added from 436 point counts in 2018. Treating all survey points as independent across years, an incidence-based iChao2 estimate (Chiu et al., 2014) of the true community size is 125 ± 2 species. As the observed metacommunity size is within the iChao2 uncertainty bounds, we retain 123 species as the true community size (*N*) used for subsequent comparisons to single-year estimates.

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We note, however, that occupancy is area-dependent, which can interact with detection probability to influence estimates of community size. In our analysis, unlimited-radius point count surveys were truncated to only include observations within 50 m of observers. To assess the sensitivity of our results to this analysis choice, all analyses were post hoc repeated in duplicate, using a detection radius of 100 m. All results pertaining to this additional analysis, which did not substantially differ, are included in Appendix S1.

2.5 | Model structure for annual diversity estimates

For each of 9 years in our empirical dataset, we separately constructed and fit a multi-species hierarchical occupancy model (Dorazio & Royle, 2005; Iknayan et al., 2014; Zipkin, Royle, Dawson, & Bates, 2010). The hierarchical structure of these models was identical to that used with our simulations (described above), with the exception that we added covariates—fit linearly via a logit-link—to help model both detection and occupancy probability (for covariate details, see Appendix S1). We augmented each year's dataset of observed species occurrences with two types of 'all-zero' species. We first included 'known' species, which was the number of species seen at some point over the 9 years that were not seen in a given year. We then added 80 'unknown' species to the dataset. Despite the different interpretation of known and unknown species, all augmented data were treated identically within the model, as all species (observed or unobserved) are treated anonymously.

We fit the model independently to each year's augmented dataset with JAGS (Plummer, 2003) using the R statistical programming language version 3.4.4 (R Core Team, 2019). In all cases, we used vague priors and hyper-priors (i.e. normal with $\mu = 0$, $\tau = 0.001$; gamma with $r = .1$, $\lambda = 0.1$). We note that a different choice of priors, following the advice of Guillera-Arroita et al. (2019), may have improved inference through increased model stability, although our results did not show strong sensitivity to choice of priors. We ran three chains in parallel with an adaptation phase of 1,000, a burn-in of 10,000 and a posterior draw of 30,000 iterations thinned by 30. Convergence of all hyperparameters and Ω was assessed visually and checked using the Gelman–Rubin statistic (Gelman et al., 2004). Following methods used for simulations, annual estimates of the true community size ($\hat{N}$), as derived separately from MSOM and iChao2, were compared to N, and performance metrics for bias, accuracy, precision and coverage probability were calculated for each (Walther & Moore, 2005).

3 | RESULTS

3.1 | Simulations

Both MSOM and iChao2 performed well across a broad range of community types. Results were similar regardless of the number of species (20 or 125) and number of sites surveyed (30 or 60). Hence, we present here the results from the scenario with 125 species and 60 sites, but provide the remainder of the results in Appendix S1. Estimates of community size were most affected by the mean occupancy probability of species in the community, regardless of the type of estimator used (Figure 1). When the mean occupancy probability of species in the community was low (0.1; scenarios 3, 6 and 7), the estimates were often biased (Figure 1a; Figure S2a), imprecise (Figure 1b; Figure S2b) and inaccurate (Figure 1c; Figure S2c) for both estimators. These effects were exacerbated when both mean occupancy and detection probability were low (scenario 6 and 7; Figure 1a), or when there was a small number of survey sites or replicate surveys (Figure 1; Figure S2). When

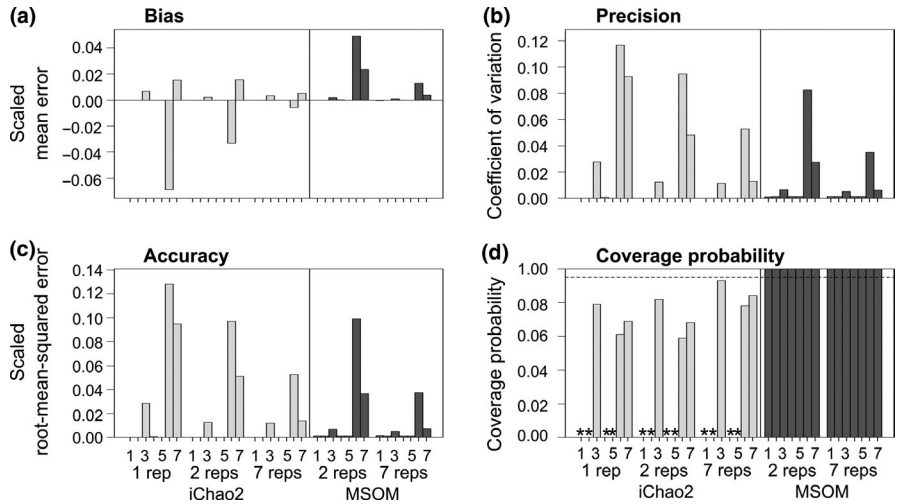


FIGURE 1 Comparison of the performance of iChao2 and multi-species occupancy model (MSOM) estimates of community size from simulations with 60 sites, 125 species, different numbers of replicate surveys ('reps' on *x*-axis), and seven different community types (listed 1–7 on the *x*-axis, each varying in mean and standard deviation of occupancy and detection; see Table 1). Each simulation was evaluated in terms of its (a) bias, (b) precision, (c) accuracy, and (d) coverage probability, relative to truth. See Figures S2 and S3 for results using both 20 species and 30 sites. We pooled the replicate surveys to determine detection or non-detection when using the iChao2 estimator with two or seven replicate surveys. The horizontal line in (d) is at 0.95. Asterisks in (d) are scenarios where estimates 95% confidence intervals are unavailable for more than 10% of simulations (see Section 2). Where applicable (a, c), performance metrics are scaled to highlight relative differences between methods

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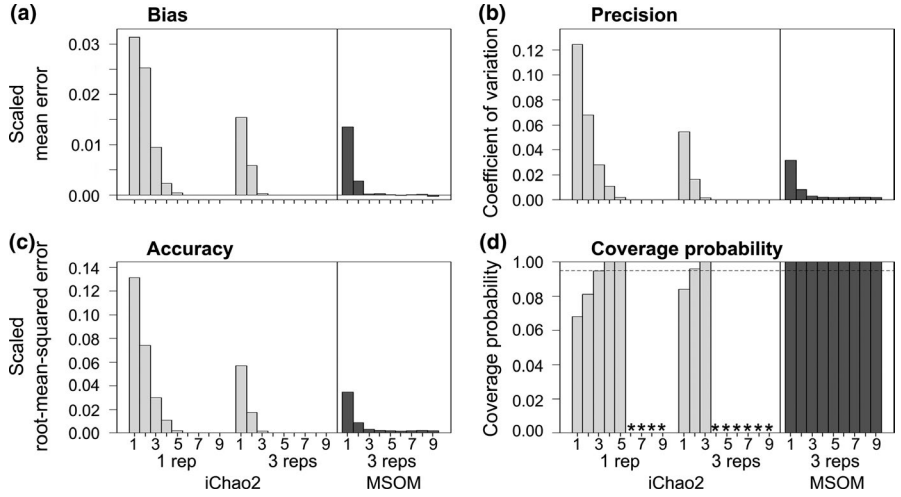


FIGURE 2 Comparison of the performance of iChao2 and multi-species occupancy model (MSOM) estimates of community size from simulations that varied the mean occupancy probability in the community from 0.1 to 0.9 by values of 0.1 (1–9 on the *x*-axis). The true community size was 50 species and we simulated three replicate surveys at 50 sites. Each simulation was evaluated in terms of its (a) bias, (b) precision, (c) accuracy, and (d) coverage probability, relative to truth. The *SD* of occupancy probabilities was 0.5. Detection probabilities of species in the community came from a normal distribution with an average and an *SD* of 0.5. The horizontal line in (d) is at 0.95. Asterisks in (d) are scenarios where estimates 95% confidence intervals are unavailable for more than 10% of simulations (see Section 2). Where applicable (a, c), performance metrics are scaled to highlight relative differences between methods

mean occupancy probability was high (0.7; scenarios 1, 2, 4 and 5), community size estimates were unbiased, precise and accurate, regardless of the estimator used (Figure 1; Figure S2). iChao2 estimates had poor coverage probability (i.e. less than 95% of the confidence intervals contained the true value; Figure 1d; Figure S3). In contrast, MSOMs had good coverage probability in all scenarios (Figure 1d; Figure S3), possibly due in part to greater uncertainty (i.e. standard error of $\hat{N}$). On average, MSOM standard errors were 5.2 species greater than iChao2 estimates using data with multiple replicate surveys (paired *t* test: $t_{(2,296)} = 53.8$, $p < .001$) and 3.1 species greater than iChao2 estimates from data with a single replicate survey (paired *t* test: $t_{(2,666)} = 25.3$, $p < .001$).

Our additional simulations to evaluate how mean occupancy probability affected estimates of community size suggested that the iChao2 estimator is more sensitive to mean occupancy probability than MSOMs, especially when only one survey replicate is used with iChao2 (Figure 2). MSOM estimates and iChao2 estimates with multiple replicate surveys are noticeably biased, imprecise and inaccurate only when the mean occupancy probability is ≤.2, given the scenarios we considered (Figure 2). However, when only one replicate survey is used with iChao2, estimates of community size are biased, imprecise and inaccurate when the mean occupancy probability is as high as .3 or .4. Furthermore, iChao2 had poor coverage probability when the mean occupancy probability was ≤.3, whereas MSOM always had good coverage probability in these simulations.

3.2 | Real dataset

Estimates of the true size of the community of terrestrial birds inhabiting post-fire forests in our focal region varied by estimation method but generally underestimated the true community size (*N*) of 123 species (Table 2; Figure 3a). Annual community size estimates ($\hat{N}$) with the iChao2 estimator varied between 84 and 134, with a median of 110, while estimates from occupancy models were generally higher, ranging between 99 and 142, with a median of 118. MSOMs were on average less biased (paired *t* test on means: $t_{(8)} = 2.1$, $p = .07$; Table 2) and trended in the direction of being more accurate (paired *t* test on means: $t_{(8)} = -1.6$, $p = .14$; Table 2).

TABLE 2 Summary of performance metrics for community size estimators, averaged across annual estimates of *N* from 9 years of empirical data on bird communities, given a presumed true *N* of 123 species. Performance evaluated in terms of bias (*ME*), precision (*Var*), accuracy (*MSE*) and coverage probability, as defined by Walther and Moore (2005). Values reported below are absolute, not relative

Performance metric	MSOM	iChao2
$\hat{N}$	116.5	108.2
Standard error of $\hat{N}$	16.5	8.8
Bias (*ME*)	−6.4	−14.8
Precision (*Var*)	162.8	200.0
Accuracy (*MSE*)	204.2	418.8
Coverage probability	1.00	0.67

Abbreviations: *ME*, mean error; *MSE*, mean squared error; MSOM, multi-species occupancy models; *Var*, variance.

Despite overall good performance, iChao2 and MSOM estimates of community size varied greatly from year to year leading to poor precision (Figure 3b). MSOM estimates had significantly larger standard errors around estimates by an average of 7.7 species (paired *t* test: $t_{(8)} = 3.7$, $p = .006$; Table 2). Larger standard errors and systematic bias, however, corresponded to better coverage, on average,

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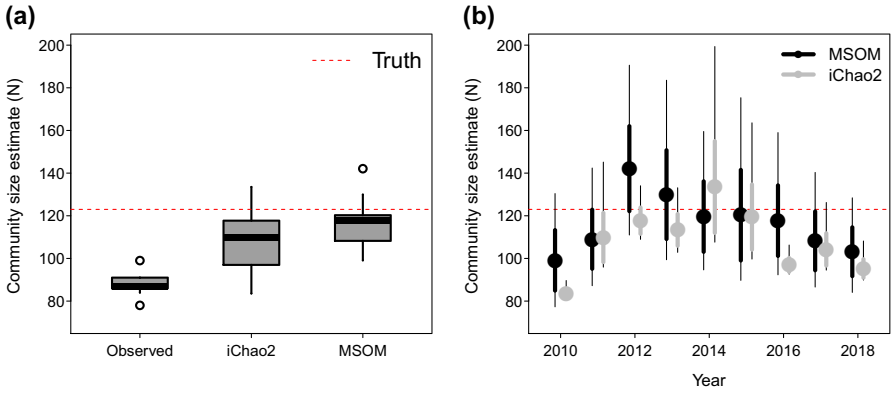


FIGURE 3 Comparisons of estimates of community size, using a real dataset comprising annual point count surveys from 2010 to 2018 for terrestrial birds within post-fire forests. With 4,190 total surveys conducted over 9 years, our knowledge of the true community size has stabilized over time, with a 9-year total of 123 species detected at least once. (a) Mean estimates of true community size across the 9 years generally underestimated the true size, but the multi-species occupancy model (MSOM) estimates showed marginally less bias. (b) Annual MSOM estimates showed wide credible intervals leading to higher coverage than iChao2 estimates. Dots show means, thick lines show means ± 1 *SD*, thin lines show 95% credible (or confidence) intervals

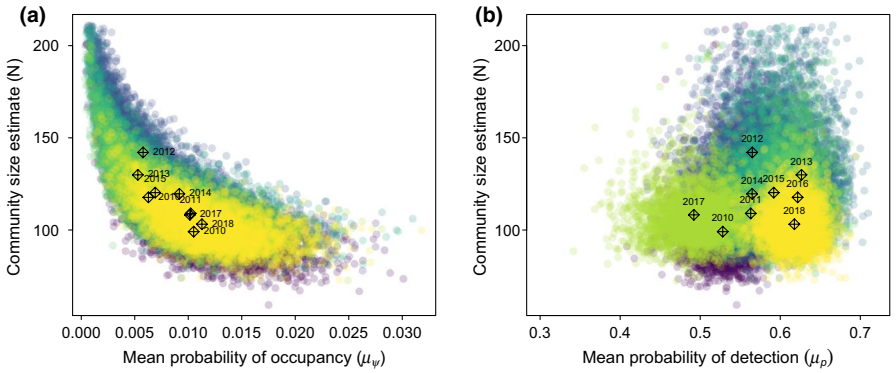


FIGURE 4 Multi-species occupancy model estimates of true community size from real data varied in response to the estimated species-wide mean probability of occupancy (μ_ψ; a) but not the mean probability of detection (μ_p; b), particularly for very low mean occupancy probabilities. In both plots, solid dots show individual posterior draws, coloured by year (e.g. dark purple = 2010; yellow = 2018), while crossed circles show labelled annual mean estimates. All plotted probabilities of occupancy and detection represent hyper-parameter values under reference conditions

with occupancy models (Figure 3b; Table 2). All nine year-specific MSOM included the true community size within 95% credible intervals, whereas only six iChao2 estimates included the true size within 95% confidence intervals. Similarly, MSOM mean estimates were within 1 *SD* of the true community size in 5 of 9 years, whereas iChao2 had similar coverage in only 3 years (Figure 3b).

Estimated community sizes from MSOM showed strong sensitivity to hyper-parameter estimates of average community-wide occupancy but not community-wide detection (Figure 4). In our dataset of real species, average occupancy probability was quite low, with a 95% credible interval across all years between 0.2% and 1.9%. In both posterior samples and annual averages, when the MSOM estimate of the mean community-wide probability of occupancy fell below approximately 0.5% (i.e. present at 0.5% of sites), then model estimates of the true community size tended to get improbably large (Figure 4a). In contrast, there was no similar relationship to detectability (Figure 4b), where community-wide mean per-survey probabilities of detection (μ_p) ranged between 46% and 66% (95% credible interval across all years).

4 | DISCUSSION

We investigated a largely unexplored feature of MSOMs that allows models to estimate the true size of regional communities, or gamma richness, via data augmentation (Dorazio & Royle, 2005; Dorazio et al., 2006; Guillera-Arroita et al., 2019). Through simulation and a case study with real data, we evaluated the performance of estimates of community size from occupancy models contrasted with the widely used non-parametric iChao2 estimator (Chiu et al., 2014). In most simulations and in our empirical case study, estimators of community size from MSOM performed at least as well, if not better, than non-parametric estimates. In simulations, iChao2 was more likely to underestimate *N* and MSOM estimates were more likely to overestimate *N*, while in the empirical data, both iChao2 and MSOM underestimated *N* on average. MSOM estimates were also more uncertain, which—when combined with bias—led to far greater coverage by MSOM estimates than iChao2 estimates in both simulations and empirical data. In all cases, precision and accuracy

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fell precipitously when the average occupancy of species was low. In general, however, our evaluations confirmed that community size estimates from multi-species occupancy models are robust, and that this extra model feature carries ecological value of its own.

4.1 | The perils of low occupancy

A persistent feature of our results is that community size estimators perform poorly when the average occupancy of all species in the community is low. In our seven simulation scenarios (Table 1), both estimators performed poorly in all scenarios where average occupancy probability was low (0.1, communities 3, 6 and 7; Figure 1). When we explored the sensitivity of estimators to average occupancy in greater detail (Figure 2), our simulations indicated a non-linear relationship between estimator performance and average occupancy—bias, imprecision and inaccuracy in community size increased substantially at the lowest values of average occupancy. Within our MSOM runs, this similar relationship appeared in our case study with real data, where individual posterior draws with very low mean occupancy probabilities showed a non-linear response (Figure 4a). This apparent non-linearity is because the relationship between estimated community size and mean occupancy probability is strongly linear on the (transformed) logit scale of the occupancy intercept hyper-parameter, $\mu_{\beta 0}$, such that a decrease in $\mu_{\beta 0}$ of 1 unit on the logit scale linearly increased the estimated community size by 29.9 ± 0.1 species (M ± SE). Consequently, uncertainty in $\mu_{\beta 0}$ leads to constant trade-offs in estimated community size that appear as non-linear when $\mu_{\beta 0}$ is transformed from the logit-scale to the probability scale.

The prevalence of low occupancy is a largely unexplored issue in MSOM (but see Broms et al., 2015). Low occupancy probability has been examined broadly in single-species studies, where precision in parameter estimates from single-species models declines non-linearly with decreasing occupancy (MacKenzie & Royle, 2005). In a multi-species context, however, low average occupancy has been observed as a phenomenon, particularly when combined with low average detectability (Si et al., 2018), but it has been otherwise little explored. Guillera-Arroita et al. (2019) explored two scenarios of MSOM performance with low average occupancy (0.27 and 0.13) but did not contrast these scenarios to those others with higher or lower occupancy. In general, low average occupancy (e.g. <0.5) should be expected for most communities, where the majority of species are uncommon to rare (Fisher et al., 1943; Preston, 1948). Particularly low average occupancy (e.g. <0.01) likely arises when study designs sample widely across biotic and abiotic conditions resulting in strong site heterogeneity, as seen with our case study where sites were spread across 2,435 m in elevation, 6.5° latitude and the first 10 years following fire where habitat is rapidly changing. Low occupancy is also highly sensitive to methodological choices—for example, when repeating our case study analysis with a detection radius of 100 m instead of 50 m, average occupancy doubled (see Appendix S1). What qualifies as 'low', consequently, will depend on how strongly the environment is driving occurrence of the average species in the community, one's choice of survey methods and whether all important environmental predictors are included in the model.

This strong habitat heterogeneity helps explain the relative difference in what constitutes 'low occupancy' between the simulations and our case study. In simulations, our lowest mean occupancy was 0.1, which resulted in poor estimator performance. In our case study, our mean occupancy was 0.01, with some posterior samples showing occupancy as low as 0.0004, but bias at 0.01 was not nearly as great as would be expected based on simulations. This apparent contradiction is likely due to several factors. First, our real dataset contained far better sampling, with a mean of 419 sites surveyed per year, whereas our simulations contained only 30 or 60 sites. Second, our real dataset included five site-specific occupancy covariates which we know strongly describe variation in occupancy probability among species inhabiting recent post-fire forests (Tingley, Ruiz-Gutiérrez, Wilkerson, Howell, & Siegel, 2016). In a study comparing non-parametric estimators of site-specific (alpha) diversity to MSOM estimates, McNew and Handel (2015) found that the inclusion of key site-specific detectability covariates made the critical difference in the superior performance of MSOM estimates. Although we did not evaluate how variation in the number (or strength) of occupancy or detectability covariates affects community size estimates, presumably, strong covariates moderate the problem of low average occupancy. Further evaluations of community size estimators would benefit from examining trade-offs in estimator accuracy and precision with variation in sample size and covariate number and strength. Meanwhile, practitioners using multi-species occupancy models should beware when using community size estimates derived from communities where average occupancy is <0.2, sample sizes are low and/or strong environmental covariates are lacking.

4.2 | Choice of community-size estimator

Ecologists have multiple methodological choices of how to estimate the true size of a given ecological community. Building on previous work (Guillera-Arroita et al., 2019; McNew & Handel, 2015), our study confirms that there are trade-offs between non-parametric estimators and multi-species occupancy models, but that both estimators perform significantly worse under particular circumstances, such as very low average occupancy. Although we did not make it an objective of our study, our results also show that either estimator is much preferable to observed counts of community size (e.g. Figure 3a), which will always be biased low due to the prevalence of imperfect detection in survey data. The choice, consequently, is not whether to use an estimator, but rather which estimator to use.

Although similar in outcome, non-parametric and MSOM methods differ strongly in their underlying assumptions and their requirements for input data. Occupancy models, in general, are data hungry, as data collected with a hierarchical structure (i.e. spatial or temporal sub-sampling) are used to independently model the occupancy and

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detectability functions (MacKenzie et al., 2002). In our simulation, for example, we were able to estimate *N* with iChao2 using either single visits, two visits, or pooling seven repeat visits (Figure 1), with remarkably similar results; all temporal sub-samples were used to estimate a single MSOM result (Figure 1). Many wildlife disciplines now commonly incorporate temporal sub-sampling into study designs, but forcing some studies—particularly citizen science or legacy datasets—into an occupancy modelling framework can violate key assumptions that lead to substantial bias (Rota et al., 2009). In these situations—and where no spatial or temporal sub-sampling was conducted—non-parametric estimators such as iChao2 hold clear and obvious advantages. While methods have been developed to apply occupancy models to single-visit study designs (Lele, Moreno, & Bayne, 2012; Sólymos, Lele, & Bayne, 2012), these methods have yet to be generalized from single- to multi-species models and their community-size estimation abilities are unknown.

The upside of the high data density required of MSOM methods is the biological realism with which the model is able to fit to the data. MSOMs can account for substantial heterogeneity in both occurrence and detection probability across sites and species, and this feature in particular has been credited for the superiority of MSOMs for estimating site-specific richness over non-parametric estimators (Dorazio & Royle, 2005; Kéry & Schmid, 2006; McNew & Handel, 2015). The appropriate modelling of substantial occupancy and detectability heterogeneity likely accounts for many of the differences between our simulations and our analysis of real data and may fully explain why the MSOM approach appears superior in the empirical analysis yet largely equivalent in the simulation. Given that other 'real-life' datasets will likely involve similar heterogeneity, the MSOM approach may provide marginally-to-substantially more accurate community size estimates than non-parametric approaches for most situations where the MSOM can be adequately applied.

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AUTHORS' CONTRIBUTIONS

M.W.T. and C.P.N. conceived the ideas; M.W.T., C.P.N. and M.E.S. designed methodology and analysed the data; M.W.T. and C.P.N. led the writing of the manuscript. All authors contributed critically to the drafts and gave final approval for publication.

DATA AVAILABILITY STATEMENT

Data and code archived as a UC Los Angeles dataset https://doi.org/10.5068/D1V097 (Tingley, Nadeau, & Sandor, 2020).

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